

Engineering Physics: BAPHY105

Few special Matrices/Operators

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Lecture Outcomes

- 1 Hermitian Operators
- 2 Unitary Operators
- 3 Projection Operators
- 4 Pauli Spin Matrices & Commutation Rules

Hermitian Operators

- It is a self-adjoint operator.

$$\hat{A} = \hat{A}^\dagger$$

- In terms of matrix representation,

$$A_{ij} = A_{ji}^*$$

- For a skew-Hermitian operator,

$$\hat{A} = -\hat{A}^\dagger$$

- ✓ Eigenvalues of a Hermitian operator are always real.
- ✓ Eigenvectors corresponding to two distinct eigenvalues of a Hermitian operator are orthogonal to each other.

Eigenvalues of a Hermitian operator are all real and its eigenvectors corresponding to two distinct eigenvalues are orthogonal.

Proof: Suppose $\hat{H}$ is a Hermitian operator.

$$\hat{H}|\chi_i\rangle = \lambda_i|\chi_i\rangle; \quad \hat{H}|\chi_j\rangle = \lambda_j|\chi_j\rangle$$

$$\therefore \langle \chi_j | \hat{H} | \chi_i \rangle = \lambda_i \langle \chi_j | \chi_i \rangle \quad (1)$$

Similarly,

$$\langle \chi_i | \hat{H} | \chi_j \rangle = \lambda_j \langle \chi_i | \chi_j \rangle \quad (2)$$

$$\Rightarrow \langle \chi_i | \hat{H} | \chi_j \rangle^\dagger = (\lambda_j \langle \chi_i | \chi_j \rangle)^\dagger \quad (3)$$

$$\Rightarrow \langle \chi_j | \hat{H}^\dagger | \chi_i \rangle = \lambda_j^* \langle \chi_j | \chi_i \rangle \quad (4)$$

Since, $\hat{H}$ is a Hermitian operator, equation-(4) becomes,

$$\langle \chi_j | \hat{H} | \chi_i \rangle = \lambda_j^* \langle \chi_j | \chi_i \rangle \quad (5)$$

From equations (1) and (5),

$$\lambda_i \langle \chi_j | \chi_i \rangle = \lambda_j^* \langle \chi_j | \chi_i \rangle$$

For $i = j$,

$$\lambda_i = \lambda_i^*$$

i.e. λ_i is real.

For $i \neq j$,

$$\langle \chi_j | \chi_i \rangle = 0$$

i.e. $|\chi_i\rangle$ and $|\chi_j\rangle$ are orthogonal to each other.

Eigenvalues of a skew-Hermitian operator are either purely imaginary or zero.

Proof: If $\hat{A}$ is a skew-Hermitian operator,

$$\hat{A} = -\hat{A}^\dagger$$

$$\hat{A}|\chi\rangle = \lambda|\chi\rangle$$

$$\Rightarrow \langle\chi|\hat{A}|\chi\rangle = \lambda\langle\chi|\chi\rangle \quad (6)$$

Taking the adjoint on both sides,

$$\langle\chi|\hat{A}|\chi\rangle^\dagger = (\lambda\langle\chi|\chi\rangle)^\dagger \quad (7)$$

$$\Rightarrow \langle\chi|\hat{A}^\dagger|\chi\rangle^\dagger = \lambda^*\langle\chi|\chi\rangle \quad (8)$$

Since, A is a skew-Hermitian matrix, equation-(8) becomes,

$$\langle \chi | \hat{A} | \chi \rangle = -\lambda^* \langle \chi | \chi \rangle \quad (9)$$

Now from equations (6) and (9),

$$\lambda = -\lambda^*$$

. Therefore λ must be purely imaginary or zero.

Unitary Operators

- The operator $\hat{U}$ is unitary if,

$$\hat{U}^\dagger \hat{U} = \hat{I} = \hat{U}^\dagger \hat{U}$$

- For an unitary operator,

$$\hat{U}^\dagger = \hat{U}^{-1}$$

- The matrix associated with the unitary operator must be invertible.
- ✓ Modulus of each eigenvalue of a unitary operator is 1.
- ✓ Eigenvectors corresponding to different eigenvalues of a unitary matrix are orthogonal to each other.

Prove that the modulus of each eigenvalue of a unitary operator is unity.

Proof: If $\hat{U}$ is a unitary operator,

$$\hat{U}|\chi\rangle = \lambda|\chi\rangle; \quad \Rightarrow \langle\chi|\hat{U}^\dagger = \langle\chi|\lambda^*$$

$$\therefore \langle\chi|\hat{U}^\dagger\hat{U}|\chi\rangle = \langle\chi|\lambda^*\lambda|\chi\rangle$$

$$\Rightarrow \langle\chi|\hat{I}|\chi\rangle = |\lambda|^2\langle\chi|\chi\rangle$$

Since, $\hat{I}|\chi\rangle = |\chi\rangle$,

$$|\lambda| = 1$$

Prove that the eigenvalues corresponding to different eigenvalues of a unitary matrix are orthogonal.

Proof: If $\hat{U}$ is a unitary operator,

$$\hat{U}|\chi_1\rangle = \lambda_1|\chi_1\rangle; \quad \hat{U}|\chi_2\rangle = \lambda_2|\chi_2\rangle$$

$$\therefore \langle\chi_1|\hat{U}^\dagger = \langle\chi_1|\lambda_1^*; \quad \langle\chi_2|\hat{U}^\dagger = \langle\chi_2|\lambda_2^*$$

$$\Rightarrow \langle\chi_1|\hat{U}^\dagger\hat{U}|\chi_2\rangle = \langle\chi_1|\lambda_1^*\lambda_2|\chi_2\rangle$$

$$\Rightarrow (1 - \lambda_1^*\lambda_2)\langle\chi_1|\chi_2\rangle = 0$$

But, for unitary operators,

$$\lambda_1^*\lambda_1 = \lambda_2^*\lambda_2 = 1$$

$$\therefore \lambda_1^*(\lambda_1 - \lambda_2)\langle\chi_1|\chi_2\rangle = 0$$

Since, λ_1 and λ_2 are distinct,

$$\lambda_1 \neq \lambda_2$$

$$\therefore \langle\chi_1|\chi_2\rangle = 0$$

i.e. $|\chi_1\rangle$ and $|\chi_2\rangle$ are orthogonal to each other.

Projection Operators

- An operator $\hat{P}$ is said to be a projection operator if it is Hermitian and equal to its own square (Idempotent).

$$\hat{P}^2 = \hat{P}$$

- For projection along a state $|u\rangle$, the projection operator can be represented as,

$$\hat{P} = |u\rangle\langle u|$$

- The operator $\hat{P}$ projects a state $|x\rangle$ along $|u\rangle$.

$$\hat{P}|x\rangle = |u\rangle\langle u|x\rangle$$

- The unit operator $\hat{I}$ is an example of a projection operator.

$$(\hat{I}^\dagger = I \text{ and } \hat{I}^2 = I)$$

Show that the operator $|\psi\rangle\langle\psi|$ is a projection operator only when $|\psi\rangle$ is normalized.

Proof: For any operator to be a projection operator, it must be Hermitian as well as Idempotent.

- Hermitian

$$(|\psi\rangle\langle\psi|)^{\dagger} = |\psi\rangle\langle\psi|$$

- Idempotent

$$(|\psi\rangle\langle\psi|)^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = |\psi\rangle\langle\psi|$$

This is possible only when $\langle\psi|\psi\rangle = 1$. This means $|\psi\rangle$ must be normalized.

Pauli spin matrices & commutation rules

- ✓ There are three Pauli spin matrices defined as,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}; \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

- ✓ Important properties of Pauli spin matrices are:

$$\sigma_x^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

Similarly, $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = I$

- $\text{Tr}(\sigma_x) = \text{Tr}(\sigma_y) = \text{Tr}(\sigma_z) = 0$
- Pauli spin matrices are Hermitian in nature, i.e.

$$\sigma_x^\dagger = \sigma_x; \sigma_y^\dagger = \sigma_y; \sigma_z^\dagger = \sigma_z$$

- Pauli spin matrices do not commute with each other.

$$\begin{aligned} [\sigma_x, \sigma_y] &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} - \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} - \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} \\ &= 2i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = 2i\sigma_z \end{aligned}$$

- Similarly,

$$[\sigma_y, \sigma_z] = 2i\sigma_x; [\sigma_z, \sigma_x] = 2i\sigma_y$$

- Pauli spin matrices anti-commute with each other.

$$\{\sigma_x, \sigma_y\} = \sigma_x\sigma_y + \sigma_y\sigma_x = 0$$

$$\{\sigma_y, \sigma_z\} = \sigma_y\sigma_z + \sigma_z\sigma_y = 0$$

$$\{\sigma_z, \sigma_x\} = \sigma_z\sigma_x + \sigma_x\sigma_z = 0$$

- In quantum mechanics the spin operator can be expressed in terms of the Pauli spin matrices as,

$$\vec{S} = \frac{1}{2}\hbar\vec{\sigma}$$